# Research on Financial Market Risk Early Warning Model Based on Multifractal Volatility and Multi-channel 1D Convolutional Neural Network Model

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**Abstract.**

This study proposes a financial risk early warning model that integrates multifractal volatility analysis with a one-dimensional multi-channel CNN framework. Using trading data from China's STAR Market Index between August 11, 2020 and January 27, 2025, and addressing class imbalance via the SMOTEENN algorithm, the model demonstrates superior performance and robustness compared to CNN-LSTM and CNN-LSTM-Attention architectures. The findings provide a novel and effective approach to financial risk prediction, with practical implications for risk prevention and regulatory decision-making.

**CCS:**

Computing methodologies~Machine learning~Machine learning approaches~Neural networks

**Keywords**:

multifractal volatility, multi-channel CNN, risk early warning, financial markets risk

## 1 INTRODUCTION

Financial market risks have intensified in recent years, as highlighted by the FDIC's 2024 Risk Review and the ECB's Financial Stability Review. Geopolitical competition, regional conflicts, and climate-related shocks have further amplified volatility across equities, bonds, commodities, and precious metals. In response, regulators worldwide have reinforced systemic risk surveillance—ranging from China's early-warning directives in the 14th Five-Year Plan to the ECB's biannual stability reports and Singapore's 2024 TRM and Cyber Hygiene notices.

Despite these efforts, forecasting financial risk remains highly challenging. GARCH-type models inadequately capture multifractal volatility, while approaches such as ARIMA and LSTM face limitations from non-stationarity, structural breaks, and overfitting. To overcome these issues, this study introduces a risk early-warning model that integrates multifractal volatility with a multi-channel convolutional neural network (CNN) to improve feature extraction and forecasting accuracy.

## 2 LITERATURE REVIEW

There are two key elements in constructing an early-warning model of financial market risk: one is the quantification of risk, and the other is forecasting of risk.

### 2.1 Quantitative Methods for Financial Market Risk

Financial risk measurement has traditionally relied on Value-at-Risk (VaR), Expected Shortfall (ES), Extreme Value Theory (EVT), and realized volatility. While VaR remains widely adopted for regulatory purposes, it ignores tail losses and underestimates diversification benefits. ES accounts for extreme losses but is highly sensitive to rare events and lacks straightforward validation. EVT improves tail modeling but suffers from threshold subjectivity and unstable

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parameter estimates, while realized volatility provides granular insights using high-frequency data but is distorted by microstructure noise and requires intensive data handling [1-3].

In recent years, multifractal volatility models based on fractal theory have emerged as powerful alternatives. Frameworks such as RV-MRW (Realized Volatility–Multifractal Random Walk), RV-BMSM (Realized Volatility–Binomial Markov Switching Multifractal), and MSM (Markov-Switching Multifractal) capture multi-scale dynamics, long memory, and self-similarity more effectively than traditional models. Empirical studies confirm their advantages: Sattarhoff and Lux (2023) showed RV-MRW and RV-BMSM outperforming ARFIMA, GARCH, and HAR across 14 international stock indices under both calm and turbulent conditions [4]. Cho and Lee (2022) integrated asymmetric Hurst exponents with GRU networks for Chinese equities, improving interval forecasts during volatility clustering [5] ; and Liu et al. (2020) demonstrated the superiority of MSM over GARCH in long-horizon forecasts for currency and equity markets [6]. Overall, multifractal volatility models provide a robust framework for capturing nonlinear, multi-scale risk dynamics.

## 2.2 Forecasting Methods for Financial Market Risk

Forecasting approaches have evolved from econometric models—such as the Kaminsky-Lizondo-Reinhart (KLR) signals, logistic regression, VAR, ARIMA, and GARCH—to machine learning methods. Classical models, though interpretable, struggle with nonlinearity, volatility clustering, and structural breaks. In contrast, machine learning, and particularly deep learning, offers stronger capacity to handle non-stationarity and high-dimensionality.

Convolutional neural networks (CNNs) are particularly effective in this domain. By extracting local features and building hierarchical representations, CNNs detect subtle temporal and spatial patterns that econometric or recurrent models may miss. Compared with RNNs, CNNs avoid vanishing gradients and are more computationally efficient, making them suitable for large-scale and real-time forecasting [7, 8]. Empirical evidence supports this advantage: Fu and Ge (2024) transformed time series into image-like matrices within a CNN framework, achieving superior predictive accuracy during stress periods [9]; Cheng et al. (2025) proposed a CNN-BiLSTM hybrid for systemic risk classification, which outperformed BiLSTM, Transformer, and TCN models in accuracy, recall, and F1-score under noisy, complex conditions [10]. These findings highlight CNNs as both effective and robust in financial risk forecasting.

## 2.3 Research Gap and Contribution

Against this backdrop, existing approaches still face challenges, including distributional misspecification, unstable tail risk estimates, and insufficient integration of multi-scale features. To address these gaps, this study develops an early-warning framework that combines multifractal volatility as a risk metric with a multi-channel CNN architecture, aiming to enhance the sensitivity and accuracy of financial market risk forecasting.

# 3 DESIGN OF A FINANCIAL MARKET RISK EARLY WARNING MODEL FOR THE STAR MARKET INDEX

## 3.1 Multifractal Volatility Estimation

Assume there are $N$ trading data points for each trading day. These $N$ data points can be divided into $n$ groups, where $n$ can take values $n_1, n_2, \ldots, n_i$, and each $n_i$ is the i-th divisor of $N$ (including 1 and $N$ itself).

The operational steps are as follows:

Step 1: The time length of each trading day is set to 1, The trading day is divided into $m$ intervals of length $\varepsilon$, where $m = \frac{1}{\varepsilon}$, and the condition $N \cdot \varepsilon = n$ is satisfied. The probability measure $P_i$ for each interval of length $\varepsilon$ within a trading day is then obtained as follows:

$$P_i(\varepsilon) = \frac{p_i(\varepsilon)}{\sum_{i=1}^{m} p_i(\varepsilon)}, i = 1,2,3,\ldots,m \quad (1)$$

Here, $p_i(\varepsilon)$ represents the sum of trading data contained within the i-th interval when the interval length is $\varepsilon$.

Step 2: Calculate the partition function $\chi_q(\varepsilon)$

According to literature [^1] and others, the following power-law relationship exists:

$$P_i(\varepsilon) \sim \varepsilon^{\alpha} \quad (2)$$

$$N_\alpha(\varepsilon) \sim \varepsilon^{-f(\alpha)} \quad (3)$$

$N_\alpha(\varepsilon)$ represents the number of intervals with the same singularity exponent $\alpha$ and interval length $\varepsilon$, while the multifractal spectrum $f(\alpha)$ denotes the fractal dimension of the subset corresponding to $\alpha$. According to literature [^2], both,$f(\alpha)$ and $\alpha$ can be calculated using the partition function $\chi_q(\varepsilon)$, which satisfies a power-law relationship, as follows:

$$\chi_q(\varepsilon) = \sum_i^m [P_i(\varepsilon)]^q \sim \varepsilon^{\tau(q)} \quad (4)$$

Step 3: Use Legendre transform to calculate the multifractal spectrum $f(\alpha)$:

$$\alpha = \frac{d\tau(q)}{dq} \quad (5)$$

$$f(\alpha) = \alpha q - \tau(q) \quad (6)$$

Through the above steps, the multifractal spectrum width for each trading day can be calculated as $\Delta\alpha = \alpha_{max} - \alpha_{min}$,where $\alpha_{max}$ and $\alpha_{min}$ represent the measure values of the intervals with the smallest and largest probability measures, respectively. A larger $\Delta\alpha$indicates a more uneven distribution of price movements during the day and greater price volatility. $\Delta\alpha$ serves as a quantitative indicator for measuring daily market volatility. Based on this approach, some scholars have developed the Multifractal Volatility Factor (MVF) as a corresponding volatility metric:

$$MVF_t = \lambda_1 \Delta\alpha_t \quad (7)$$

$$\lambda_1 = \frac{D^{-1}\sum_{t=1}^{D} R_t^2}{D^{-1}\sum_{t=1}^{D} \Delta\alpha_t} \quad (8)$$

$\Delta\alpha$ only utilizes two recorded values of $\alpha$ within a single day (i.e., the maximum and minimum), which does not fully capture the entire range of probability variations in price fluctuations, and thus cannot adequately reflect the true magnitude of price volatility. Moreover, the distribution of $\Delta\alpha$ is highly sensitive to extreme values. Therefore, some scholars have proposed using the standard deviation of $\alpha$ values—denoted as $S_\alpha$ —when constructing multifractal volatility measures.

$$MVM_t = \lambda S_{\alpha,t} \quad (9)$$

$$\lambda = \frac{D^{-1}\sum_{t=1}^{D} RV_t}{D^{-1}\sum_{t=1}^{D} S_{\alpha,t}} \quad (10)$$

Here,

$$RV_t = \eta RV_t^* \quad (11)$$

$$RV_t^* = \sum_{i=1}^{m} R_{t,i}^2 \quad (12)$$

## 3.2 Multi-channel 1D CNN Architecture

The target of the model prediction is a risk label, which is a discrete variable taking only the values 0 or 1. This study employs a multi-channel one-dimensional convolutional neural network (1D CNN) for prediction. The fundamental building block of the multi-channel 1D CNN model is the single-channel 1D CNN.

Unlike traditional CNN architectures that typically feature a single input layer and output layer, the multi-channel CNN extracts features from different input signals independently (through convolution, pooling, etc.),

then concatenates the resulting high-dimensional features. The fused feature map is subsequently fed into fully connected layers for final output. This approach enables the multi-channel 1D CNN to not only extract deep features from each individual channel, but also to implicitly capture the topological relationships between different input signals. In the context of structural vibration response signals, this facilitates more in-depth mining and extraction of damage-related information. The architecture of the multi-channel 1D CNN network is shown in Figure 1.

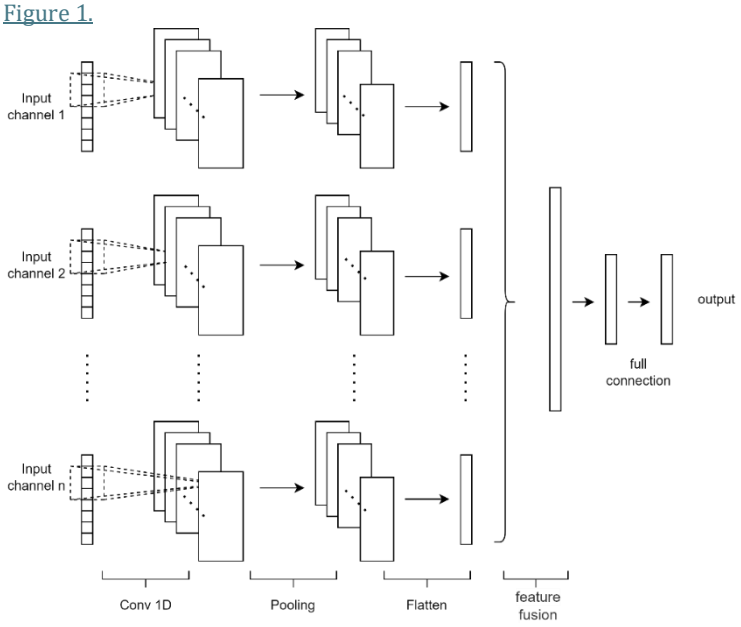



**Figure 1.** **Structure of The Multi-Channel 1D CNN Model**

In the multi-channel 1D CNN used for financial market risk early warning, multiple parallel input channels are employed to extract different features from trading data. Each channel performs layer-by-layer extraction of high-dimensional features through convolution and pooling operations. The extracted features are then fused and concatenated, followed by fully connected layers to produce the final risk classification output.

# 4 EMPIRICAL STUDY

This study selects trading data of the STAR Market Index (000688) from August 11, 2020 to January 27, 2025 for multifractal feature analysis, comprising a total of 1,085 trading days. The data are sourced from the WIND database.

## 4.1 Verification of Multifractal Characteristics

To balance data volume and prediction accuracy as much as possible, this study conducts multifractal analysis using the 5-minute closing prices of the STAR Market Index on a daily basis. The interval lengths for non-overlapping and evenly spaced partitions of the 48 data points per trading day are set to 1, 1/2, 1/3, 1/4, 1/6, 1/8, 1/12, 1/16, 1/24, and 1/48. The range of $q$ is determined based on the saturation values of $\alpha$ and $f(\alpha)$, and through calculation, $q$ is set to the interval [-5,5]. Four trading days (September 23, 2020; October 19, 2020; November 24, 2022; and March 8, 2023) are randomly selected to test for multifractal characteristics, and both the multifractal spectrum distribution curves and the double-logarithmic plots of $\ln\chi_q \sim \ln\varepsilon$ are generated.

As shown in Figure 2, first, the relationship between the multifractal spectrum $f(\alpha)$ and $\alpha$ is a unimodal function, indicating that the series exhibits multifractal characteristics. Second, when $q$ takes different values, the curve $\ln\chi_q \sim \ln\varepsilon$ displays a good linear relationship, suggesting the presence of multifractal features in the closing price series of the STAR Market 50 Index. In empirical studies, data from other trading days were repeatedly selected, and the corresponding multifractal spectrum $f(\alpha)$ curves and the $\ln\chi_q \sim \ln\varepsilon$ curves consistently exhibited similar characteristics. This demonstrates that the intraday price fluctuations of the STAR Market 50 Index show significant multifractal behavior, thereby laying a solid foundation for identifying financial market risk states of the index using multifractal methods in this paper.

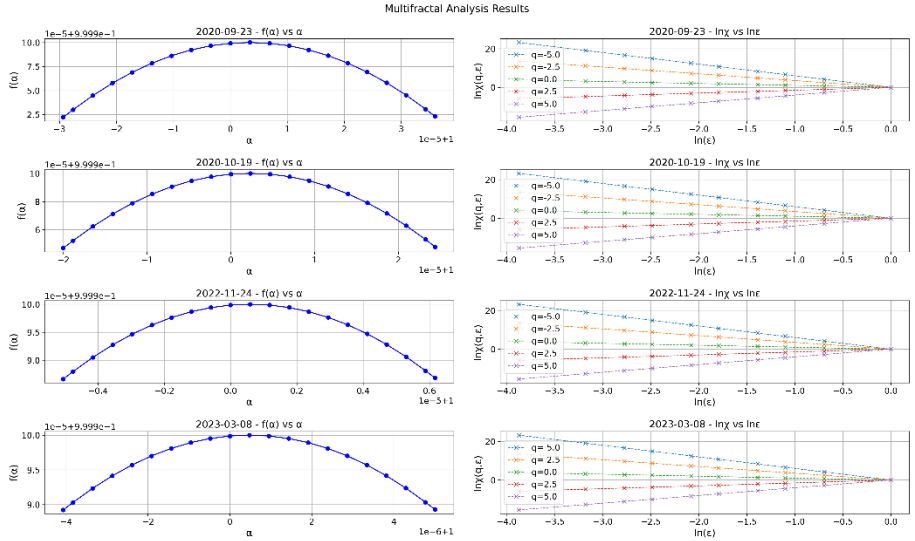



**Figure 2.** **Multifractal Characteristics**

## 4.2 Classification of Daily Risk Categories for the STAR Market Index

To determine the state variable for each sample, it is essential to define whether the internet financial market is in a risk state on a given day. In this study, the multifractal volatility series $MVM_t$ is constructed by calculating the standard deviation $S_\alpha$ of the daily closing price's $\Delta\alpha$ and the scaled realized volatility $RV_t$. Figure 3 presents the daily return series$R_t$ and the multifractal volatility series $MVM_t$ over the sample period.

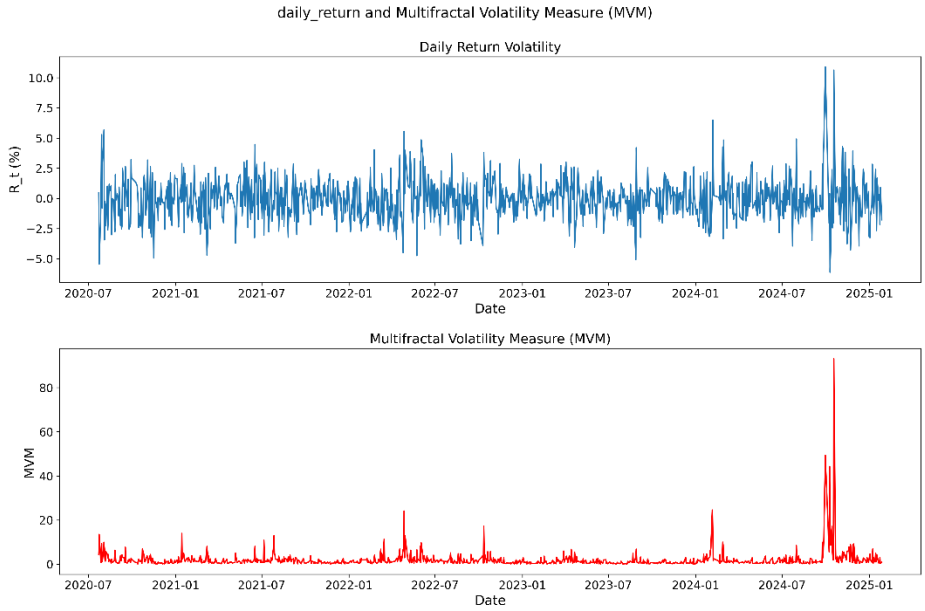



**Figure 3.** **Daily Return Series and The Multifractal Volatility Series**

If, during a certain period, both the daily return $R_t$ and the multifractal volatility $MVM_t$ are significantly higher than those in other periods, it indicates that the intraday price fluctuations of the STAR 50 Index during that time are relatively large, implying that the financial market is experiencing higher risk. Zhao Haiyue and Yang Wanshou (2019) selected the bottom 10% of the total sample as financial risk samples and achieved favorable research outcomes. Following this approach, this study selects the bottom 10% of samples with the lowest $R_t$ values or the top 10% of samples with the highest $MVM_t$ values as "risk-concern" samples of the internet financial market, based on the calculated results of $R_t$ and $MVM_t$. Out of the 1,085 trading days of the STAR Market Index, a total of 103 are classified as "risk-concern" samples, while the remaining 982 are labeled as "normal" samples.

Normality tests show that both $R_t$ and $MVM_t$ do not follow a normal distribution. Therefore, the Kolmogorov–Smirnov (K–S) test is applied to both variables. The results reject the null hypothesis at the 1% significance level, indicating that both $R_t$ and $MVM_t$ significantly distinguish between normal and risk samples. Consequently, the classification of normal and risk-concern samples based on $R_t$ and $MVM_t$ significantly distinguish between normal and risk samples. Consequently, the classification of normal and risk-concern samples based on $R_t$ and $MVM_t$ is statistically valid, accurate, and reliable.

## 4.3 Selection of Feature Indicators for the Risk Early Warning Model

To construct a risk early warning model for the STAR Market (Sci-Tech Innovation Board), it is essential to accurately identify the market feature indicators that drive the emergence of financial risks. Drawing on relevant literature (Wang Peng & Huang Xun, 2018; Xu Xuechen & Tian Kan, 2021), this study initially selects eight basic indicators of the STAR Market Index—opening price, closing price, highest price, lowest price, previous closing price, rate of change, return, trading volume, and trading amount. In addition, six technical indicators are included: three stochastic indicators [KDJ(K), KDJ(D), KDJ(J)], the Williams %R (WR), moving average (MA), and

on-balance volume (OBV). All of the above data are sourced from the WIND database. Descriptive statistics for these indicators are provided in Table 1.

**Table 1. Descriptive Statistics of Candidate Indicators**

| Index | mean | max | min | sdv | skew | kurt |
|---|---|---|---|---|---|---|
| open | 1093.00 | 1630.30 | 648.13 | 252.22 | 0.22 | -1.06 |
| close | 1092.24 | 1610.78 | 643.04 | 251.48 | 0.22 | -1.05 |
| high | 1105.21 | 1639.19 | 654.16 | 254.42 | 0.22 | -1.05 |
| low | 1080.76 | 1584.31 | 635.74 | 248.65 | 0.21 | -1.07 |
| pre | 1092.78 | 1610.78 | 643.04 | 251.80 | 0.22 | -1.05 |
| ROC | -0.76 | 98.71 | -73.32 | 18.08 | 0.19 | 1.78 |
| Return | -0.03 | 17.88 | -6.13 | 1.82 | 2.02 | 17.31 |
| trade_volume | 138.39 | 811.03 | 30.74 | 87.72 | 2.27 | 9.21 |
| trade_amount | 5473.2 | 27716.76 | 1685.02 | 2816.09 | 2.38 | 10.06 |
| KDJ(K) | 45.36 | 96.43 | 1.9 | 27.46 | 0.17 | -1.35 |
| KDJ(D) | 45.37 | 94.3 | 4.03 | 25.32 | 0.17 | -1.31 |
| KDJ(J) | 45.33 | 115.78 | -15.42 | 36.3 | 0.15 | -1.3 |
| WR | -55.45 | 0 | -100 | 31.21 | 0.19 | -1.3 |
| MA10 | 1094.67 | 1588.21 | 654.58 | 251.79 | 0.21 | -1.09 |
| OBV | -1572.78 | 504.58 | -4172.32 | 1150.65 | 0.02 | -1.06 |

Note: The statistical values of trading volume, trading amount, and the OBV indicator (excluding skewness and kurtosis) are expressed in units of ten million.

Data source: WIND Database.

The mutual information values of each candidate indicator were calculated, and the results are shown in Figure 4.

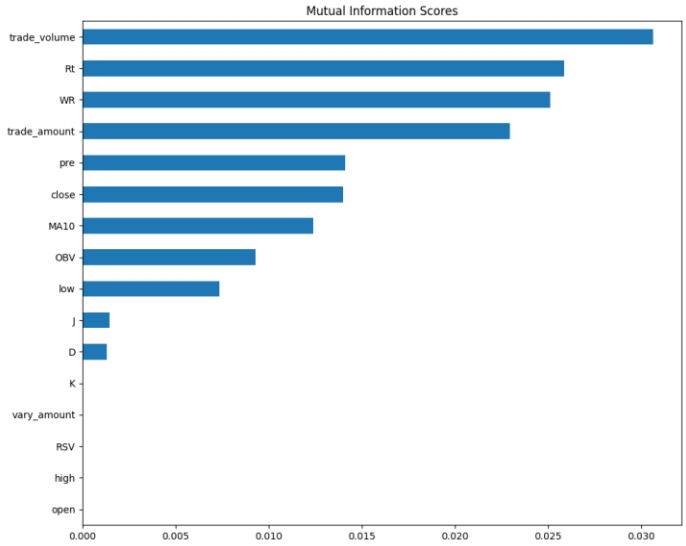



**Figure 4. Mutual Information Score**

The features with mutual information (MI) values greater than zero—namely, trading volume, return, Williams %R (WR), trading amount, previous closing price, closing price, 10-day moving average (MA10), On-Balance Volume (OBV), lowest price, KDJ(D), and KDJ(J)—were selected as the predictors for the risk label in this study.

## 4.4 Model Architecture and Parameter Settings

The model is built using TensorFlow. Table 2 presents the architecture and key parameter settings of the multi-channel 1D CNN model. In the input, convolutional, and pooling layers, all 11 input channels share the same structure. In the first convolutional layer, the number of filters is set to 32 with a kernel size of 3. The pooling size is uniformly set to 2, the activation function used is ReLU, and the dropout rate is 0.3. The extracted feature maps are then merged through a Concatenate layer, followed by two fully connected layers with 64 and 1 units respectively, using ReLU and Sigmoid as activation functions.

Similarly, the single-channel 1D CNN retains only one input channel, while the rest of the architecture remains the same. The initial learning rate of the model is set to 0.001, and the Adam optimizer is used to dynamically adjust the learning rate. The maximum number of training epochs is set to 10. Cross-entropy loss is used as the training objective, and the sliding window step size is set to 10.

**Table 2. Model Configuration and Parameter Setting**

| Layer | Type | Parameter Setting |
|---|---|---|
| Input | InputLayer | shape=(10,1) |
| conv | Conv1D | (32,3,'relu') |
| drop | Dropout | 0.3 |
| pool | MaxPooling1D | pool_size=2 |
| flat_k | Flatten(·) | -- |
| … | … | … |
| flat_merge | Concatenate | flat_k(k=1,2,…,11) |
| dense_1 | Dense | (64,'relu') |
| dense_2 | Dense | (1,'sigmoid') |

Note: The architectures of the 11 input channels are identical, i.e., $k = 1, 2, \ldots, 11$.

## 4.5 Model Training

1. Train-Test Split

Based on the sample dataset constructed from the aforementioned feature indicators and state variable, there are 109 "risk concern" samples and 988 "normal" samples, indicating an imbalanced dataset. To evaluate the model, the dataset is split into training and testing sets. The training set accounts for 70% of the total samples, including 76 risk concern samples and 691 normal samples. The test set makes up the remaining 30%, consisting of 33 risk concern samples and 297 normal samples.

2. Handling Imbalanced Samples

The Synthetic Minority Over-Sampling Technique combined with Edited Nearest Neighbors (SMOTEENN) is applied to the training set to generate synthetic samples for the minority (risk concern) class. These synthetic samples are combined with the original training data to form a new, balanced training set. After data cleaning, the resulting training set contains 875 risk concern samples and 981 normal samples, compared to the original 76 and 691, respectively.

3. Overfitting Prevention

When training neural networks, the model's performance on the training set typically improves with more training epochs. However, performance on the validation set may begin to degrade after a certain point, indicating

overfitting—where the model learns the training data too well but fails to generalize to unseen data. To prevent overfitting and retain the best-performing model on the validation set, this study adopts an early stopping strategy.

The validation loss is used as the monitoring metric, with a patience parameter of 10 epochs. This means that if the validation performance does not improve over 10 consecutive epochs, training will be terminated early. Consequently, model training may stop before reaching the maximum number of epochs if no further improvement is observed within the patience window following the best validation result.

## 4.6 Performance of the Financial Risk Warning Model

1. Multifractal Volatility-Based Determination Results of Risk Trading Days

Figure 5 shows the trading days identified as risk days by the multifractal volatility method. It can be observed that risk trading days are mainly concentrated in three periods.

The first period spans 2020 to 2021 during the outbreak of the COVID-19 pandemic. The development of the pandemic brought tremendous uncertainty to the global economic outlook. First, there was a liquidity shock. After the COVID-19 outbreak in 2020, the Federal Reserve rapidly cut interest rates to stimulate economic recovery and implemented a series of unconventional monetary policies, including large-scale purchases of government and corporate bonds. Meanwhile, China's COVID-19 policies enabled it to resume work and production earlier, boosting market optimism and driving the index to historic highs. However, by the end of 2021, as the economy recovered, the Federal Reserve began discussing tapering its accommodative monetary policies. This policy shift significantly impacted market liquidity, leading to a sharp decline in the STAR 50 Index at the end of 2021. Second, profit growth slowed. After 2021, the earnings growth rate of the STAR 50 began to decline, which affected market confidence and investor decision-making to some extent. The combined shocks to fundamentals and liquidity caused continuous adjustments in the STAR Market after 2021, further intensifying market volatility.

The second period is February 2022, when the Russia-Ukraine conflict erupted. The escalating global turmoil, combined with repeated COVID-19 outbreaks and slowing economic growth, further increased market uncertainty and intensified the volatility of the STAR 50 Index. In March of the same year, the Federal Reserve initiated a new round of interest rate hikes, which added pressure on China's cutting-edge industries such as semiconductors and artificial intelligence. After a recovery and rise in the index, a rapid decline followed, causing severe turbulence.

The third period is late September 2024, when the People's Bank of China announced a re-lending policy lowering the re-lending interest rate. Expectations for Fed rate cuts grew stronger, and the long-dormant market sentiment surged unprecedentedly. However, due to the still-slow macroeconomic growth, the market had not fully recovered, and starting from October, a partial correction occurred. The trading market became extraordinarily active, with investor behavior causing intense volatility.

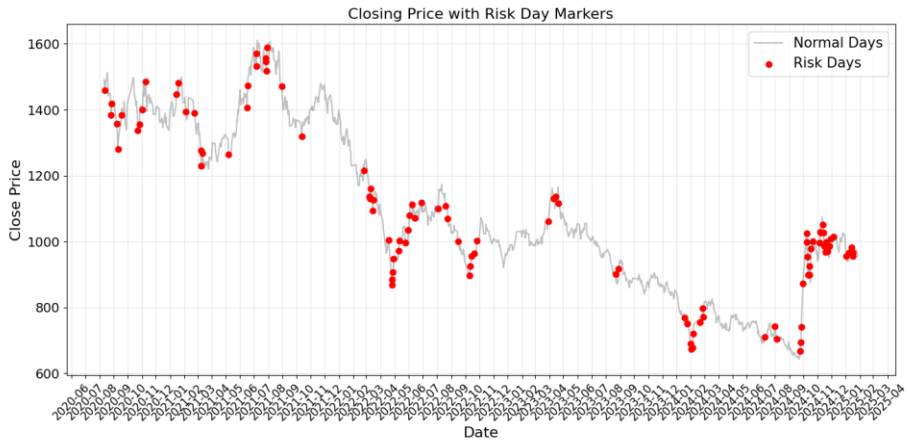



**Figure 5.** Determination Results of Risk Trading Days

2. Prediction Performance of the Risk Early Warning Model

Considering that the early warning model in this study is based on a multi-channel CNN, which is essentially a binary classification model, the F1-score is chosen as the primary metric to comprehensively evaluate model performance. Additionally, accuracy, precision, recall, and ROC curves are also provided as supplementary references.

The F1-score is calculated using the following formula:

$$F1 - Score = \frac{2 \times accuracy \times recall}{accuracy + recall} \quad (13)$$

Here,

$$accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (14)$$

$$recall = \frac{TP}{TP+TN} \quad (15)$$

Additionally,

$$precision = \frac{TP}{TP+FP} \quad (16)$$

Table 3 and Figure 6 respectively present the evaluation metrics , confusion matrices and ROC curves for the single-channel CNN model, multi-channel CNN model, CNN-LSTM model, and CNN-LSTM-Attention model.

The results show that:

1. CNN models outperform LSTM models.
2. Multi-channel CNN performs better than single-channel CNN.
3. LSTM with an attention mechanism outperforms the standard LSTM.

**Table 3** Prediction performance metrics of different models

| models | accuracy | precision | recall | f1 |
|---|---|---|---|---|
| **multiCNN** | 0.985915 | 0.989619 | 0.980571 | 0.985075 |
| **CNN_LSTM** | 0.771939 | 0.970954 | 0.534857 | 0.689757 |

| models | accuracy | precision | recall | f1 |
|---|---|---|---|---|
| CNN_LSTM_Attention | 0.861322 | 0.99839 | 0.708571 | 0.828877 |

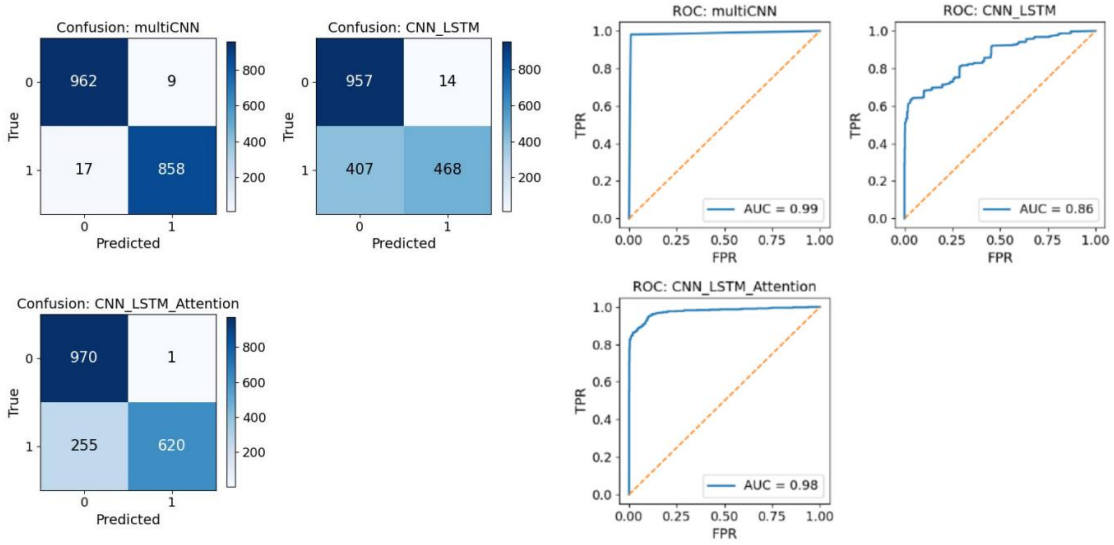



**Figure 6.** Confusion Matrix and ROC Curve

3. Robustness Test

To verify the robustness of the constructed early warning model, this study conducts a comparative analysis of the model's performance metrics by varying the size of the training dataset. Results are shown in Table 4.

**Table 4** Performance Comparison of Models in Robustness Tests with the New Training Set

| model | Training Set Sampling Ratio | accuracy | precision | recall | F1 |
|---|---|---|---|---|---|
| Multi-CNN | 25% | 0.9740 | 0.9859 | 0.9589 | 0.9721 |
| | 50% | 0.9648 | 0.9868 | 0.9383 | 0.9619 |
| | 75% | 0.9778 | 0.9793 | 0.9737 | 0.9765 |

From Table 4, it can be seen that after changing the training set sample size, the performance metrics of the Multi-CNN model did not exhibit significant fluctuations. The mean values remained stable and the standard deviations were small, indicating that the model is robust and possesses reliable stability. It can effectively predict risks in China's internet financial market, providing a useful operational tool for financial regulators and investors to manage risk.

# 5 CONCLUSION

This study focuses on the STAR 50 Index to construct a multi-channel CNN model for early warning of financial market risks in China. First, based on fractal theory, the multifractal characteristics of the STAR Market Index are revealed. Then, multifractal returns are used to characterize the financial risk states of the index and classify risk levels, with feature indicators accurately extracted through statistical tests and principal component analysis. To address the imbalance issue in samples, the hybrid sampling technique SMOTEENN is incorporated into the multi-channel CNN model, effectively improving the traditional multi-channel CNN's prediction performance on

imbalanced data. The prediction performance of the multi-channel CNN model is compared and analyzed in terms of prediction accuracy, significance, and robustness. The main conclusions are as follows:

First, China's STAR Market Index financial market exhibits multifractal features. Multifractal volatility captures the complex characteristics of market fluctuations, accurately characterizes internet financial risk states, and reasonably distinguishes between normal and risk-concern states, effectively reflecting the real conditions of China's internet financial market.

Second, through analysis of the driving factors of internet financial risk, 14 candidate indicators were initially identified. Mutual information calculations were then used to select 11 feature indicator variables.

Third, the multi-channel CNN early warning model constructed in this study shows significant advantages in predicting financial risks of the STAR Index. In terms of prediction accuracy and robustness, this model outperforms not only the CNN-LSTM model but also the CNN-LSTM-Attention hybrid model that incorporates an attention mechanism. This demonstrates that the multi-channel CNN model possesses excellent predictive performance for financial risk forecasting in China's STAR Market Index.

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